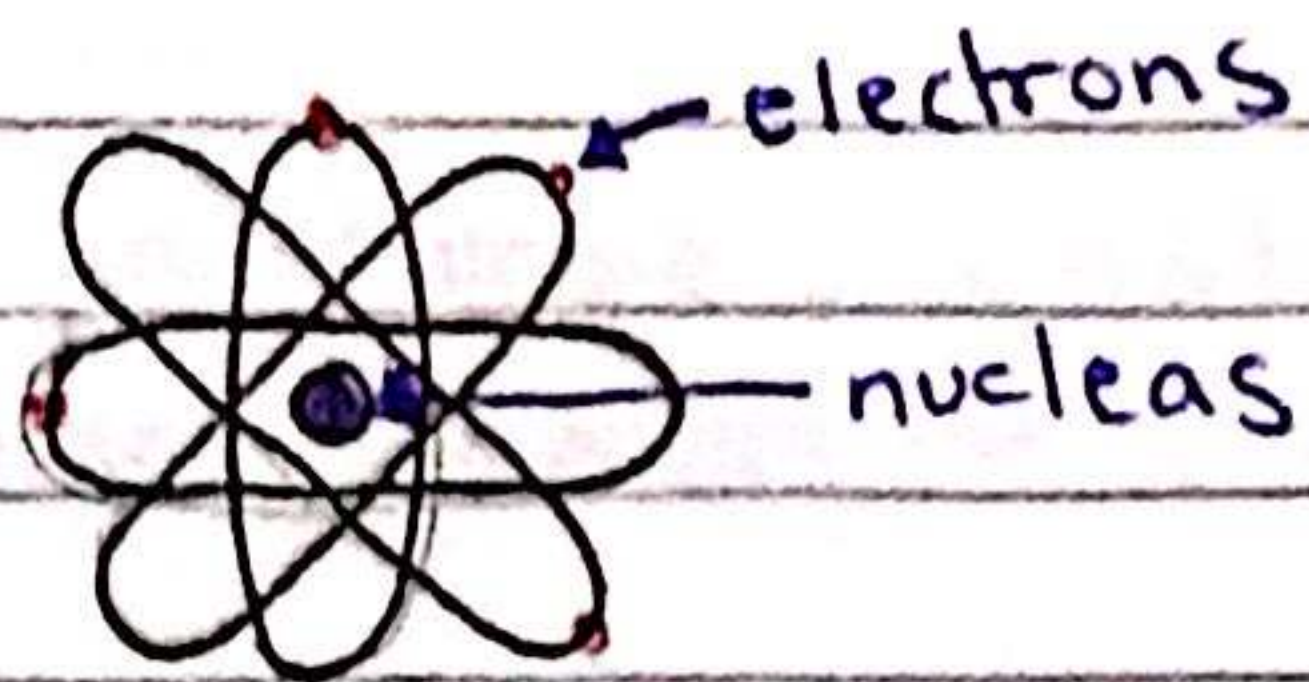
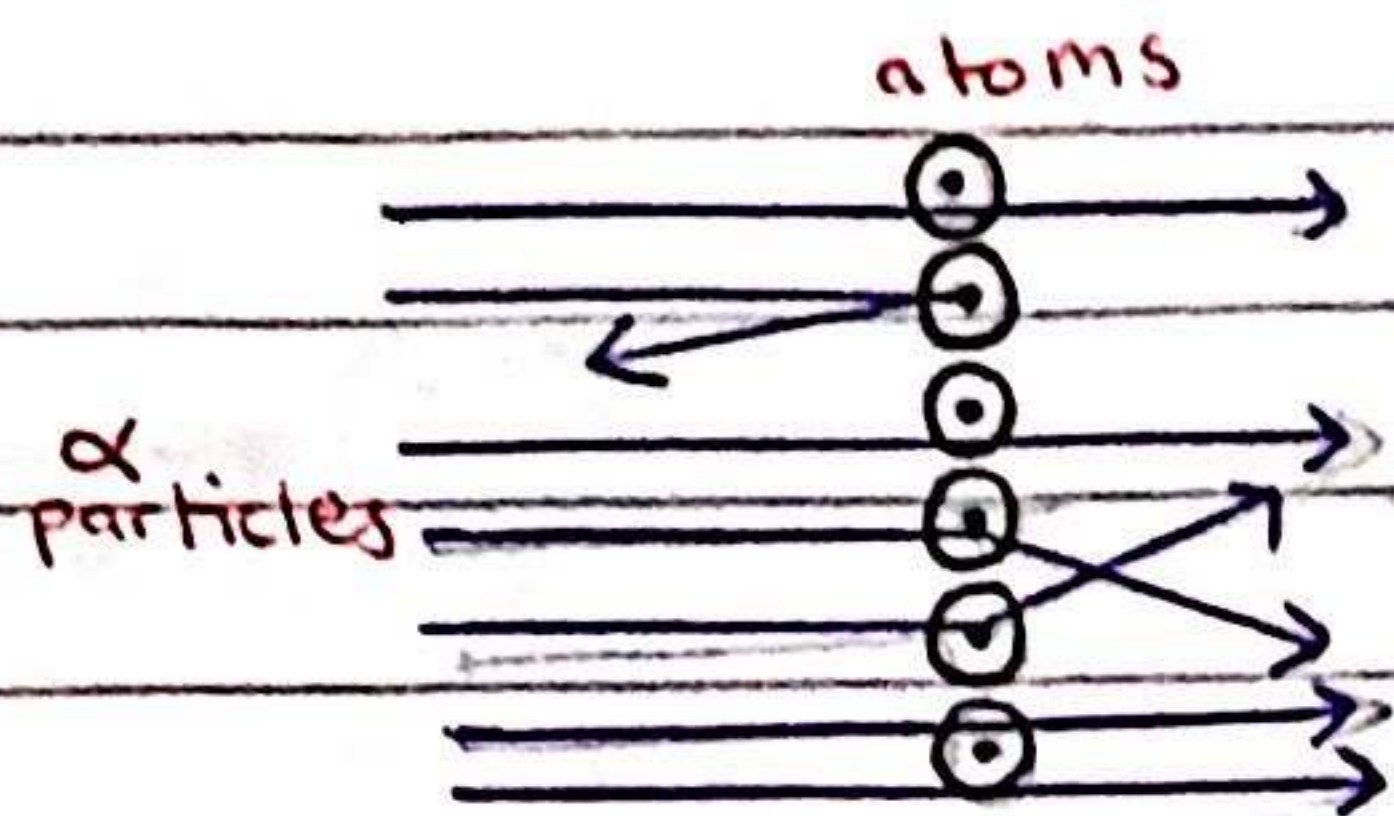


Atom :-

• the atom is
x100 000 larger
than the
nucleus

- if an atom loses electrons, it becomes a positive ion and vice versa

Rutherford's experiment :-

- majority of the lines go straight
↳ because the atom is mainly empty space
- some are deflected through small angles
↳ the positive α particles repel the positive nucleus which contains most of its mass
- a very small number are deflected straight back
↳ because the nucleus is extremely small

* the nucleus consists of protons & neutrons

- electrons have a -1 charge while protons have a $+1$ charge and neutrons have 0 charge
- an atomic mass is the proton number & the mass number is the sum of protons & neutrons

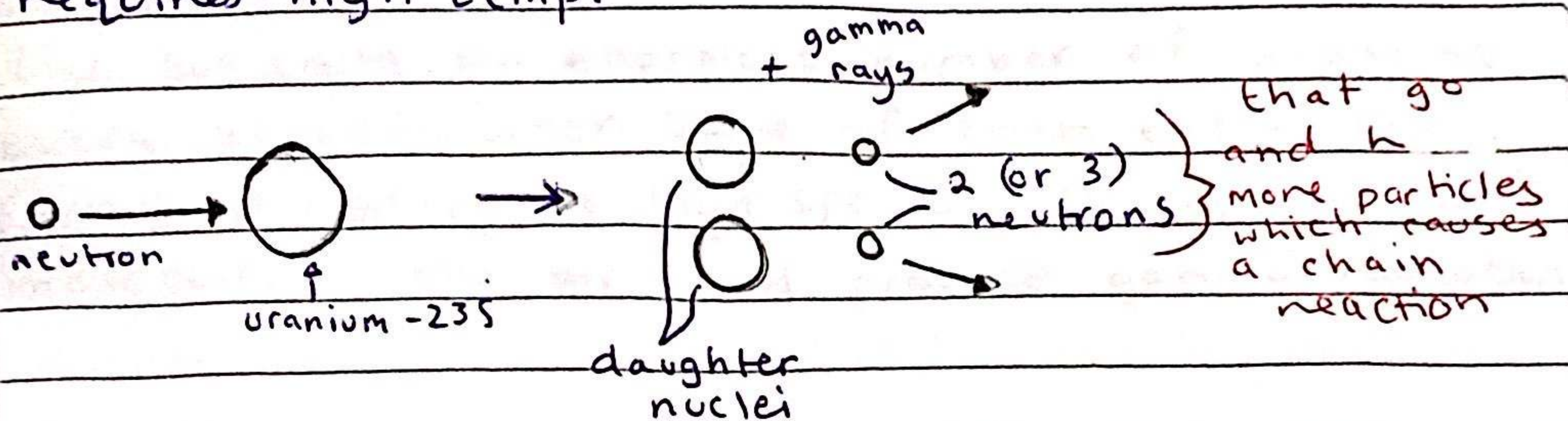
* an isotope is different atoms of the same element with the same number of protons but a different number of neutrons

Nuclear Fission:

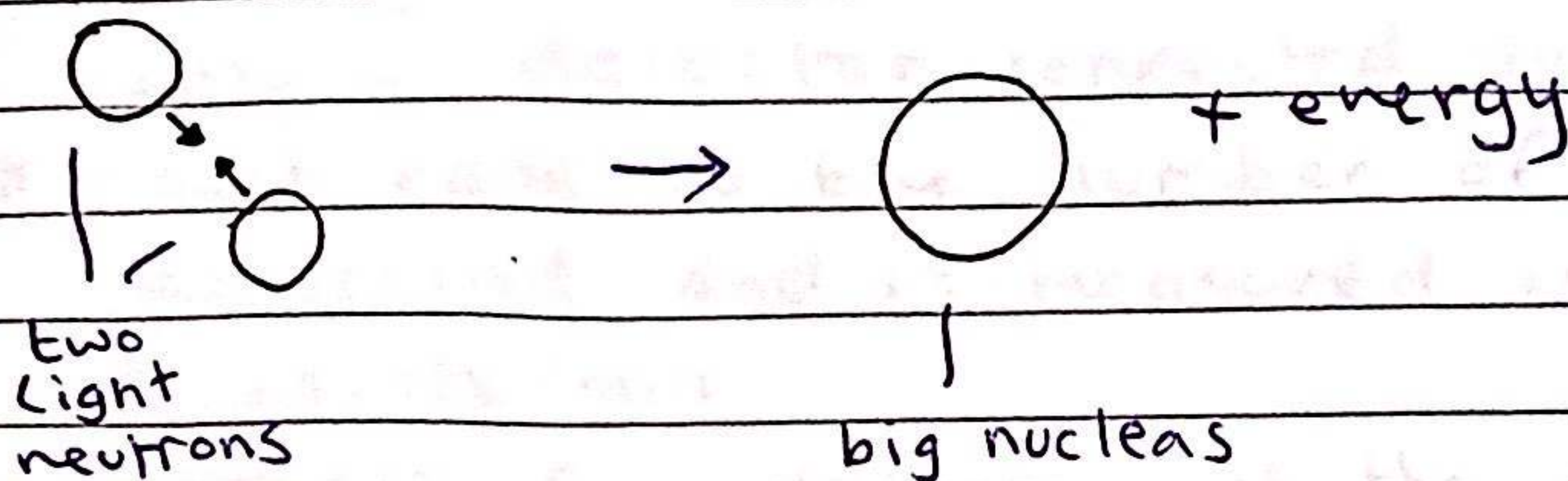
- the splitting of a large, unstable nucleus into two smaller (daughter) nuclei

Nuclear Fusion:

- when two light nuclei join to form a heavier nucleus
- * requires high temp.



Fission



Background radiation:

- radiation is a natural phenomenon
- background radiation is radiation that exists around us all the time

the sources

radon gas:

- from the natural decay of uranium in rocks & soil
- tasteless, colorless & odorless

04/01/23

rocks and buildings:

natural radioactivity can be found in building materials, including rocks, stone & brick

Food and drink:

naturally occurring radioactive elements can get into food and water

cosmic rays:

the sun emits an enormous number of protons every second, when some of them enter the Earth's atmosphere at high speeds, they collide with molecules in the air and produce gamma radiation

* ionising nuclear radiation can be measured using a detector connected to a counter

* count rate is the number of decays per second and is measured in counts/s OR counts/min

* example of a detector is the Geiger-Müller tube

Nuclear emission:

emission of radiation from a nucleus is spontaneous and random in direction.

types of emissions

alpha (α), beta (β), gamma (γ)

particle	charge	penetration	ionization
alpha (α)	+2	stopped by paper	high
beta (β)	-1	stopped by aluminum	medium
gamma (γ)	0	reduced by lead	low

electromagnetic wave

- a particle is deflected in an electric field if it has charge
- a particle is deflected in a magnetic field if it has charge & is moving perpendicular to it
- therefore
 - γ particles are not deflected by either
 - α & β particles are deflected

- ionising effects depend on: $\alpha > \beta > \gamma$
- electric charge
greater the charge more ionising it is
 - kinetic energy
higher the E_k , more ionising it is

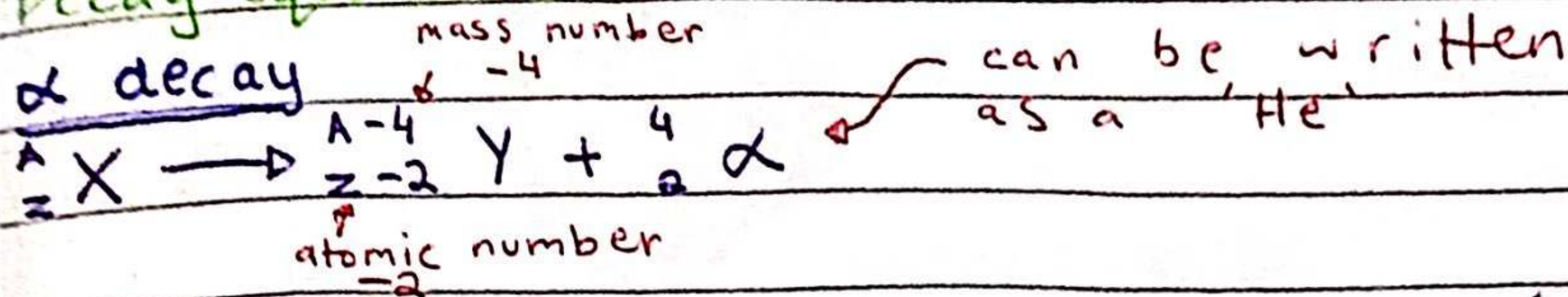
Radioactive decay:

- a change in an unstable nucleus that can result in the emission of α particles or β particles and/or γ radiation
 - all of this is spontaneous & random
- * during an α/β decay, the nucleus changes to that of a different element
- * isotopes of an element can be radioactive due to an excess of neutrons in the nucleus and/or the nucleus being too heavy
- * a nucleus decays to increase its stability by reducing the number of excess neutrons
- ↳ this is done by the α and β decay
 - ↳ if the nucleus has too much energy, it is given off, often in the form of gamma (γ) radiation

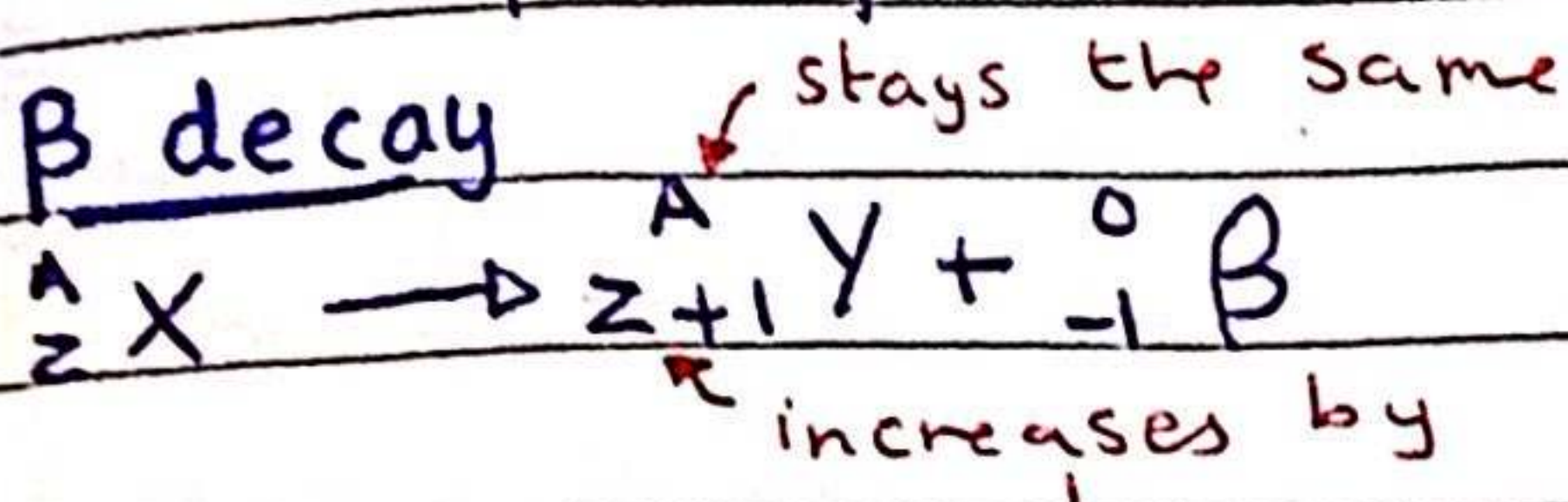
- β emissions

↳ neutron $\rightarrow$ proton + electron

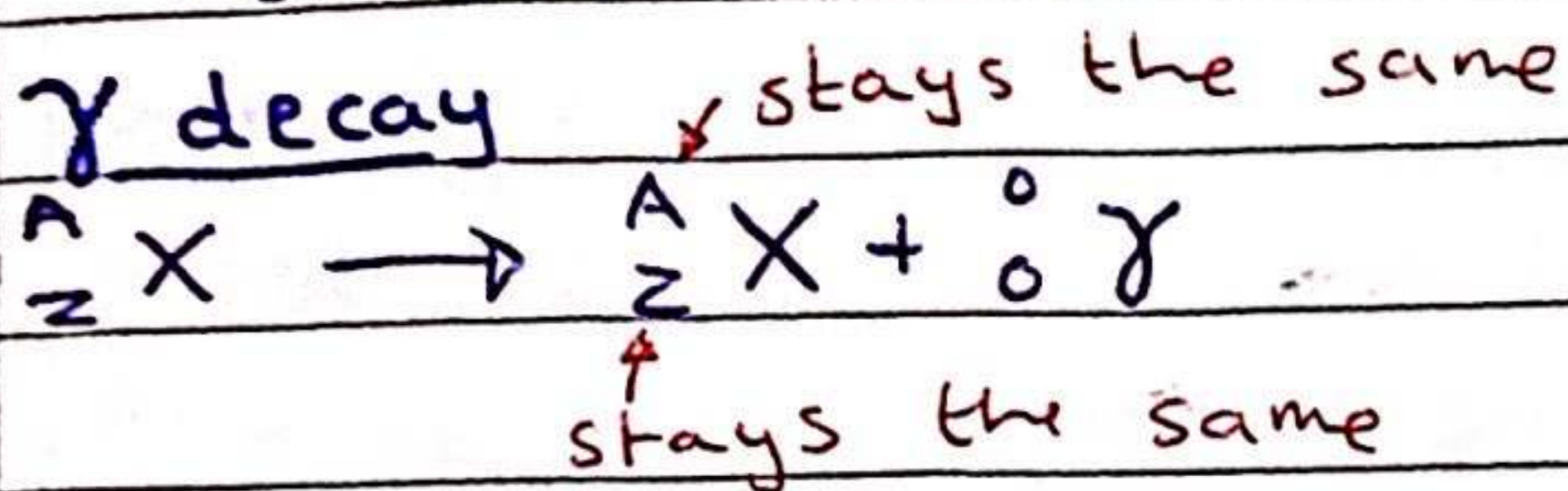
Decay equations:



• an alpha particle is emitted ($\alpha = \text{'He'}$)



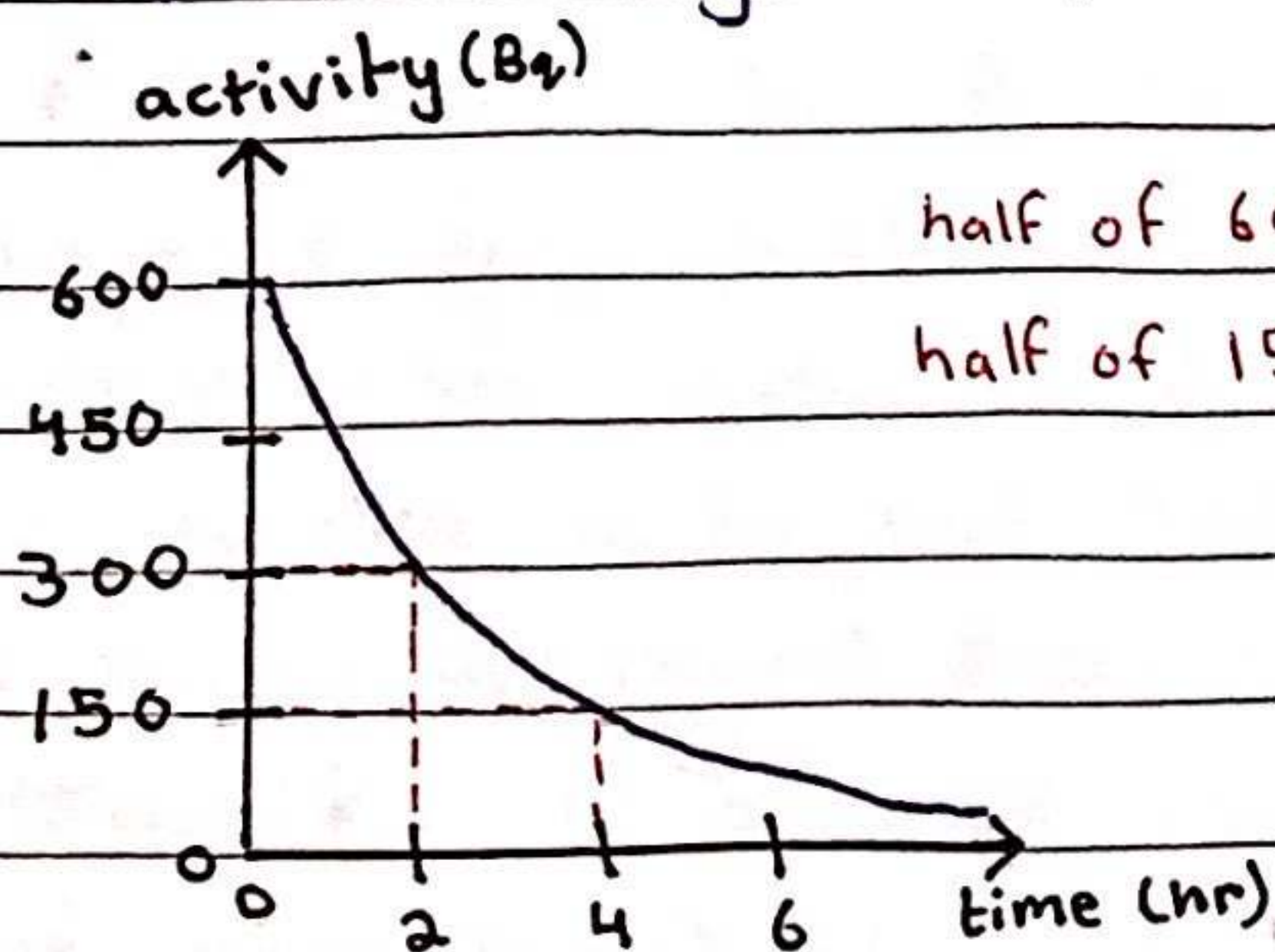
• a neutron changes into a proton + electron where the electron is emitted and the proton remains in the nuclei.



• the gamma ray emitted has a lot of energy but no mass or charge

Half-life:

• the time taken for half the nuclei of that isotope in any sample to decay

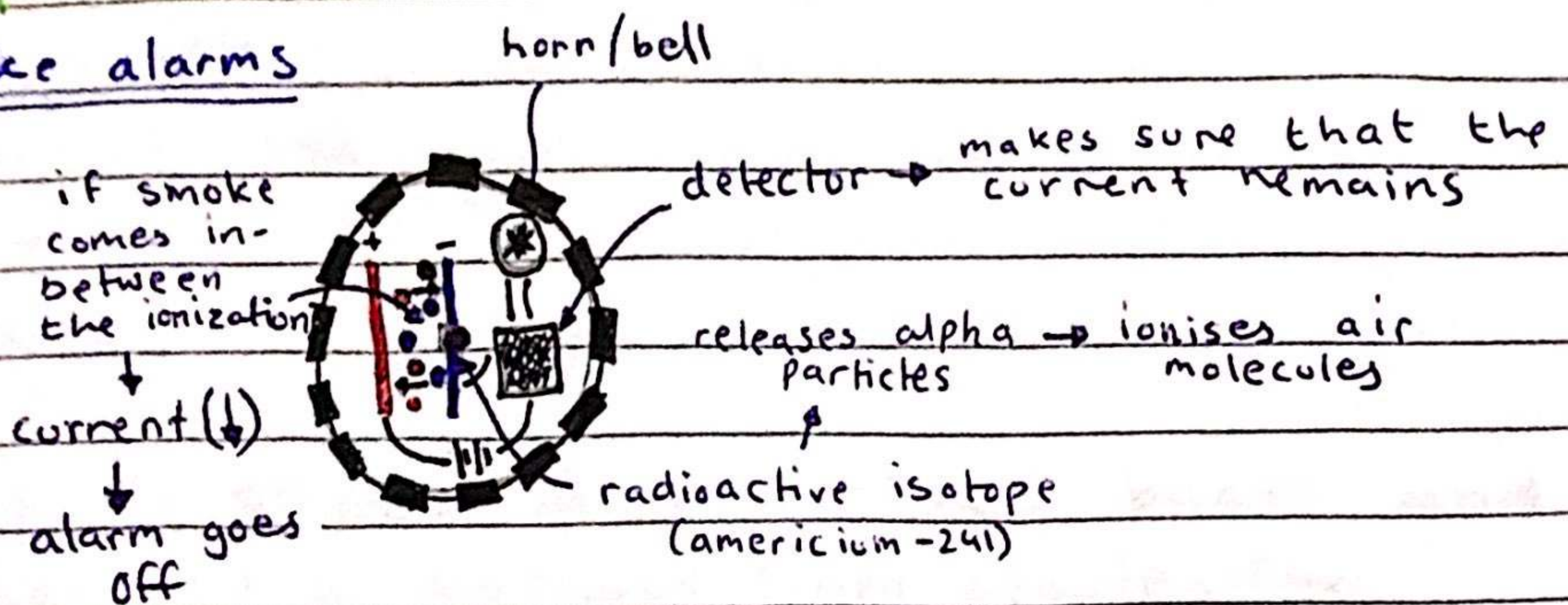


half of 600 = 300 $\rightarrow$ 2 hr

half of 150 = 75 $\rightarrow$ 2 hr

Applications:

Smoke alarms



Sterilising food & medical equipment

- gamma radiation is widely used to sterilise medical equipment
→ because it's the most penetrating
- gamma radiation can also be used to kill micro-organisms in food

measuring thickness of materials

- most commonly, beta particles are used
- the amount of beta particles that penetrate through can be monitored by a detector
- thicker the material, less particles get through and vice versa
- this allows machines to make adjustments for a constant thickness
- ' α ' would all be absorbed & ' γ ' would all get through so ' β ' is the best option

diagnosis and treatment of cancer

- gamma rays are used because they can penetrate the human body and reach the tumor
- the beams are moved around to minimise harm to tissues
- a 'tracer' is a radioactive isotope that tracks the movement of substances like blood.
- a PET scan detects the emissions from the tracer to diagnose cancer and find the tumour's location.

Effects of ionising nuclear radiation:

- cell death
- tissue damage
- mutations
- cancer

* it is stored in lead-lined boxes and kept at a distance from people. The amount of time it is used should be minimal and should be used from as far as possible or by using tongs or other equipment; gloves, safety goggles ...

- reduce exposure time
- increase distance between source & living tissue
- use shielding to absorb radiation